Robustness of spectral methods in community detection

Ludovic Stephan[1,2], Laurent Massoulié[1,3]

[1]Inria Paris, École Normale Supérieure, PSL Research University

[2]Sorbonne Université

[3]Microsoft Research-Inria Joint Centre

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Background & setting

Community detection / Graph clustering: finding community structure in graphs

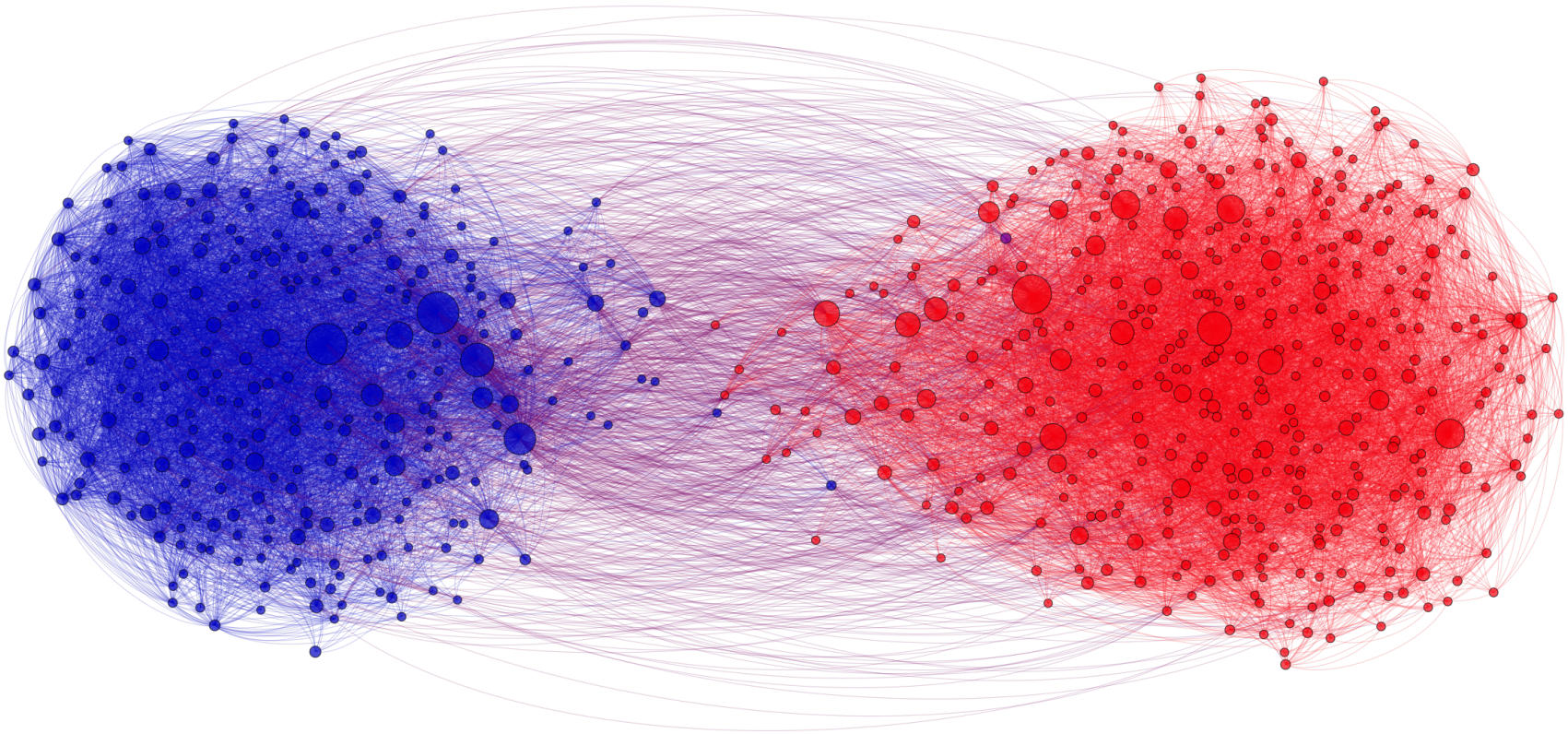


Figure: Example of graph with community structure

Stochastic block model

Generative model to test efficiency of clustering algorithms

- n vertices,
- community labels $(\sigma(i))_{i\in[n]}$ uniformly chosen in $[r]$,
- random graph G with vertex set $\{1,\dots,n\}$ and edge probabilities

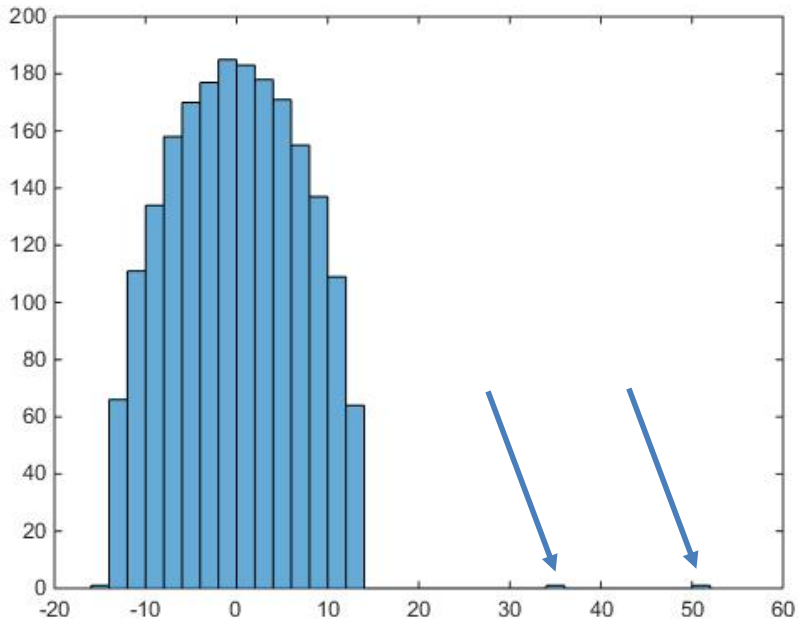
$$\mathbb{P}(\{i,j\}\in E) = \frac{W_{\sigma(i)\sigma(j)}}{n}$$

for some fixed symmetric $r\times r$ matrix W.

Goal: Recover the community labels from G

Spectral methods

Idea: study the spectrum of a matrix M associated with G (usually the adjacency matrix).
When the signal is strong enough, the top eigenvectors of M are correlated with the community structure, while the rest is confined in a "bulk".



Sparse case

Problem: In our case (edge probabilities in $O(1/n)$), not enough signal !

However, other methods have been found:

- self-avoiding matrix [Mas13]: two communities,
- non-backtracking path counts [MNS13]: idem
- non-backtracking matrix [KMM+15], [BLM15]: requires an asymmetry condition on the eigenvalues of W.

Both only work under the Kesten-Stigum condition:

$$\mu_2^2 > \mu_1,$$

where $\mu_1 > |\mu_2| > \dots$ are the eigenvalues of W.

New matrix for the sparse case

We introduce the matrix $D^{(\ell)}$, defined by

$$D_{ij}^{(\ell)} = 1 \quad \text{iff} \quad d(i,j) = \ell$$

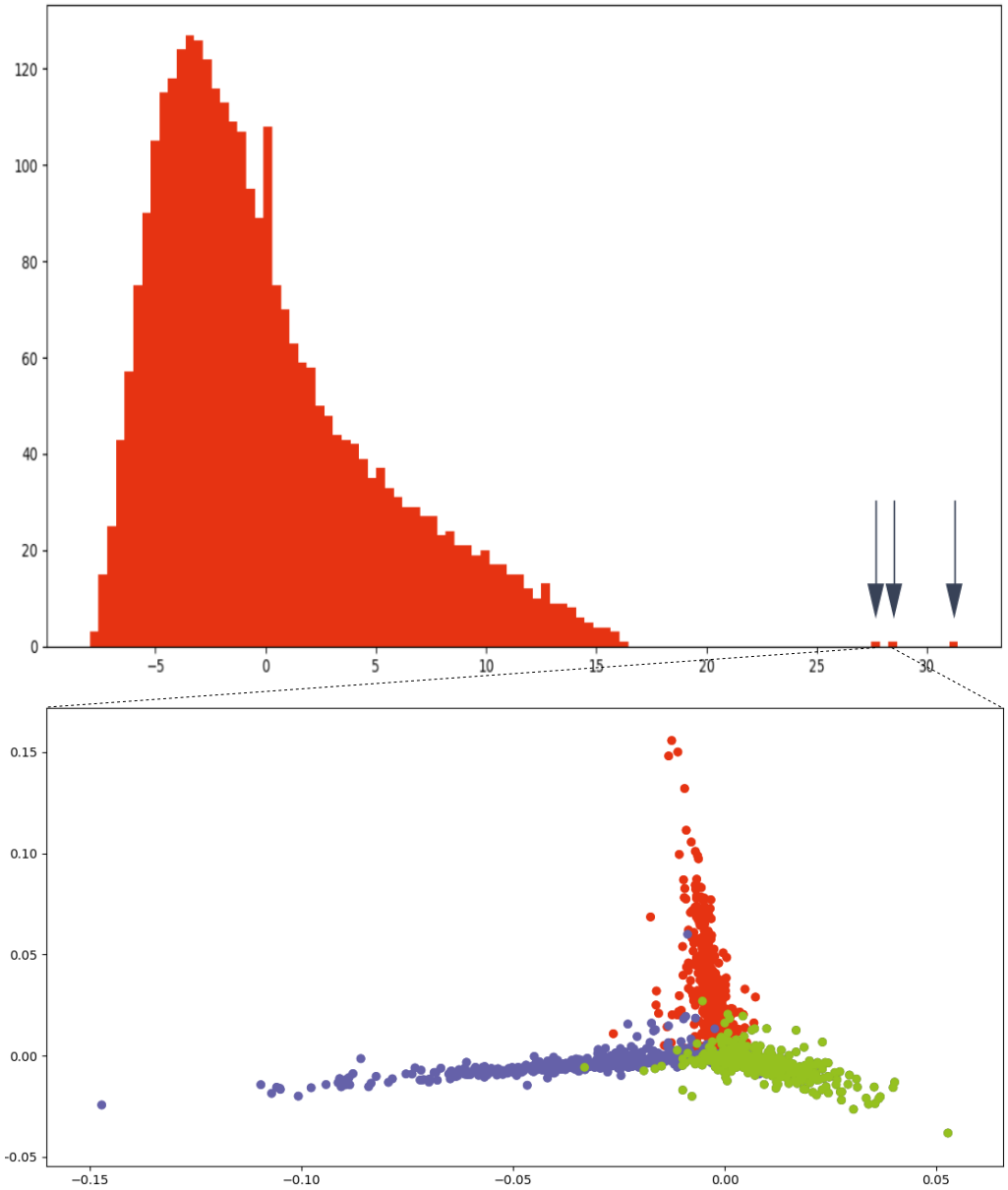
$\Longrightarrow$ generalization of the adjacency matrix when $\ell > 1$.

Theorem

Under the Kesten-Stigum condition, when $\ell = c \log(n)$ for small enough c, the matrix $D^{(\ell)}$ exhibits the desired spectral structure

No asymmetry condition, works for any number of communities !

Spectral structure of $D^{(\ell)}$



Robustness of spectral methods

Real-life networks differ from SBMs: more small cliques/denser subgraphs, independently of community structure
$\Longrightarrow$ need to study the robustness to perturbations

Adversary definition

An adversary of strength γ can arbitrarily add and remove edges in G, as long as at most γ vertices are affected.

Already known spectral methods (non-backtracking, path expansion) are known to be extremely sensitive to this adversary.

Stability of $D^{(\ell)}$

The matrix $D^{(\ell)}$ is sparser the the previous ones $\Rightarrow$ less sensitive to noise !

Theorem

Let $\tau = \mu_2^2/\mu_1$. Then, when $\ell = c\log(n)$ for small enough c, the distance matrix $D^{(\ell)}$ retains the same spectral structure as long as

$$\gamma = O(\tau^\ell/\log(n))$$

This allows perturbations of size n^ε, for ϵ approaching zero as we near the KS threshold.

Discussion

Comparison with SDP methods, commonly used for robustness.

- less stable: SDP allows perturbations of size $O(n)$
- more efficient: our algorithm runs in $O(n^{13/12})$, which is fast !
- we reach the KS threshold, while SDP methods can only come close to it as the degree increases.